



# Ryan Zernach Full-Stack AI Systems Engineer • Agentic Infrastructure & Product Engineering

Portfolio: [Ryan.Zernach.com](https://ryan.zernach.com) • Email: [Ryan@Zernach.com](mailto:Ryan@Zernach.com) • [LinkedIn](#) • [GitHub](#) • [Remote](#) • [FinTech](#)

## HIGHLIGHTS

- Distributed systems engineer with 10+ years of engineering Payments, Agents, Mobile Apps, Websites, DevOps + MLOps Systems
- Scaled CellarChat™ tool-calling agent to 1M+ active members, complete with RLHF & CI/CD LLM evaluation scoring pipelines
- Engineered Apple Pay subscriptions to lift recurring revenue by 25%+ which lifted mobile app ratings to 4.9★ (10k+ reviews)
- Completed Gauntlet AI engineering fellowship with 2% acceptance rate and mastered the art of agentic AI orchestration
- Designed observable backend systems that support autonomous AI-powered VoIP eCommerce payment automations
- Awarded Top 10 of 10,000 international projects in 2024 NASA Space Apps hackathon with AI Voice Assistant

## TECH STACK

### Programming Languages

TypeScript, Python, Rust, Golang, Java, C#, Ruby, Swift, Kotlin, Bash, JavaScript, PHP, C++, Objective-C, R, Solidity, Dart, Flutter, XML

### AI Systems & Machine Learning

LLMOps, vLLM, LiteLLM, RAG, Agents, MCP, Vector Embeddings, Evaluations, Semantic Kernel, LangChain, Langfuse, Computer Vision, PyTorch, Knowledge Graphs, TensorFlow, Scikit-Learn, Bedrock, Foundry

### Platform & Backend Engineering

DevEx, Node.js, FastAPI, Django, Nest.js, Express, .NET Core, OpenAPI, Microservices, REST, GraphQL, WebSockets, Webhooks, gRPC, Kafka

### Data & Persistence

PostgreSQL, SQL, NoSQL, Redis, Vector Databases, ETL, Snowflake, Graph Databases, Entity Relationships, Caching, BigQuery, PySpark

### Cloud Infrastructure & DevOps

AWS, Azure, GCP, Cloudflare, Docker, GitHub Actions, Terraform, Kubernetes, CI/CD, CloudOps, Blue/Green, DNS, Networking

### Front-End & Product Engineering

React, React Native, iOS, Android, Web, Swift, Kotlin, Flutter, Expo Router, Next.js, Redux, Streamlit, Animations, Design, User Feedback, Playwright

### Distributed Systems & Operations

Queues, Redis, RabbitMQ, Analytics, Pub/Sub, Linux, Airflow, VoIP, Cron, WebRTC, Observability, OpenTelemetry, Rate Limiting, Horizontal Scaling, Machine Learning CI/CD, Event-Driven Architecture, Load Balancing

### Security, Identity & Compliance

AuthN, AuthZ, OAuth, SOC 2, Identity, Role-Based Access Control, JWT, Secure Systems Design, SSO, API Security, Secrets Management

### Payments, Communications & Integrations

PCI Compliance, Stripe, Apple Pay, IAP, Billing, Subscriptions, Wallets, Twilio, SMS, Push Notifications, HTTP APIs, Marketing Tech, SMTP/IMAP

### Blockchain & Web3

Ethereum, Solana, dApps, WalletConnect, MetaMask, Polygon, Solidity, Smart Contracts, Ethers.js, Tokens, ERC-20, ERC-721

## EXPERIENCE

### Independent Full-Stack AI Systems Engineer • USA • Remote

July 2025 - Present

- Building agentic products and back-end systems across Python, Typescript, vector embeddings + OSS large-language models
- Automated eCommerce marketplace responsibilities using Twilio VoIP, realtime voice agents + DALL-E image generations
- Dockerized scalable Robotic Process Automation (RPA) via Kubernetes, local vLLM inferences, and agentic web-scraping
- Published TranslateWallpaper, a cross-platform mobile app to make LLM-powered language-learning wallpapers images

### Senior Full-Stack AI Systems Engineer • CellarTracker • Seattle • Remote

Aug 2023 - Jul 2025

- Scaled agentic CellarChat™ to 1M+ international members & built world's highest-rated #1 wine app
- Architected RLHF pipeline to fine-tune OpenAI's GPT-4o, OpenAPI CI/CD triggers, embeddings, & evals
- Integrated ML wine recommendations using Azure-ML + world's largest online database of wine reviews
- Maintained Python, TypeScript, React Native, C#/.NET, SQL, Redis, CI/CD, Azure AI Foundry + PromptFlow
- Helped to boost app store ratings to 4.9★ average with 10,000+ verified 5-star ratings on iOS and Android
- Embedded Apple Pay subscriptions in the React Native app to generate additional 25%+ recurring revenue
- Developed backend Python systems supporting agentic workflows, support tickets, & account management
- Improved user's experience of cellar management CRUD workflows to improve consistency and reduce taps
- Implemented PyTorch OCR image-to-text search, high-performance barcode scanning, and activity feed posts
- Streamlined the Redux store for fast, efficient, and persistent global state management between app sessions
- Rebuilt the React Native app's analytics implementations for tracking critical user interactions and in-app behavior
- Arranged A/B tests during paywall & onboarding experiences using phased rollout schedules for marketing analysis
- Migrated 25+ year old legacy support ticket system from Active Server Pages (ASP) to TypeScript Express framework

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| <b>Full-Stack Mobile Engineer</b> • Archlife Industries Software • Miami • Remote                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Sept 2021 – Aug 2023</b>  |
| <ul style="list-style-type: none"> <li>• Productionized stocks/crypto exchange to compete with Robinhood: public leaderboards + copy friend's portfolio distribution</li> <li>• Used AWS Terraform to deploy docker container running, Java + Springboot, &amp; NodeJS dashboards, and SQL migrations</li> <li>• Constructed core mobile app functionality, including: Plaid connections, trending investors, &amp; suggested stocks</li> <li>• Built social network features, such as direct chat, user profiles, posts, likes, emojis, &amp; threaded comments</li> <li>• Wrote the data-visualization logic for viewing and comparing the performance of different stocks/crypto</li> <li>• Transitioned eSports React Native codebase from JavaScript to TypeScript for long-term maintainability</li> <li>• Improved Ruby on Rails for push notifications, chat, &amp; native calendar sync to improve user satisfaction</li> <li>• Remedied hardware's bluetooth transactions in Swift telecommunications app + Python Flask app</li> </ul> |                              |
| <b>Data Science + Machine Learning Tutor</b> • edX • Vanderbilt, UC Berkeley, +24 others • Remote                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>July 2020 – Sept 2021</b> |
| <ul style="list-style-type: none"> <li>• During pandemic, taught 900+ online college students Python, PyTorch, APIs, machine learning, TypeScript, SQL + databases</li> <li>• Managed busy schedule of remotely teaching 30+ university students per week in 1:1 setting (one hour per person)</li> <li>• Dozens of universities were subscribed to my employer's curriculum and online tutoring resources for students</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                              |
| <b>Full-Stack React Web Developer</b> • Georgia Hybrid Landscaping • Alpharetta, Georgia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Oct 2015 – Jan 2019</b>   |
| <ul style="list-style-type: none"> <li>• Engineered uploading videos/images, survey questions, pricing proposals, &amp; processing payments via Java + Spring Boot</li> <li>• Migrated websites from PHP to NodeJS React, which successfully processed \$500k+ gross sales in three years</li> <li>• Managed online eCommerce payments, visitor analytics, Google AdSpend, and email marketing campaigns</li> <li>• Configured automated, outbound marketing email campaign schedulers &amp; listeners using IMAP/POP</li> <li>• Setup observability dashboards for payments, marketing, analytics, API logs, &amp; admin</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                              |                              |
| <b>Payments Systems Intern (Card Processing)</b> • Cost Plus Processing • Alpharetta, Georgia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Sept 2012 – Jan 2013</b>  |
| <ul style="list-style-type: none"> <li>• Worked with merchant processing systems &amp; payment gateways to understand PCI compliance and financial flows</li> <li>• Analyzed end-to-end card transaction lifecycles (interchange rates, risk) across critical payment networks</li> <li>• Built foundational knowledge of secure payment infrastructure, interchange fees, &amp; transaction routing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                              |

## EDUCATION & TRAINING

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| <b>AI Engineering Fellowship (G3)</b> • Gauntlet AI • Remote • Austin, TX                                                                                                                                                                                                             | <b>2025</b> |
| <ul style="list-style-type: none"> <li>• Mastered how to command agents, automate workflows, and 100x enterprise-grade software engineering</li> <li>• Harnessed the power of RAG, multi-agent systems, OpenAPI, and MCP to build agentic automations</li> </ul>                      |             |
| <b>Data Science &amp; Machine Learning Bootcamp</b> • Lambda School • Y Combinator • Remote                                                                                                                                                                                           | <b>2020</b> |
| <ul style="list-style-type: none"> <li>• Recommendation Engines, Computer Vision (CV) Models, Time-Series Forecasting, Natural-Language Processing</li> <li>• Python, SciKit-Learn, PySpark, Django, Flask, FastAPI, Data Engineering, PostgreSQL Databases, Back-End APIs</li> </ul> |             |
| <b>Turfgrass &amp; Landscape Management</b> • University of Georgia (NALP) • Athens, Georgia                                                                                                                                                                                          | <b>2015</b> |
| <ul style="list-style-type: none"> <li>• Design Principles, Science of Grass, Outdoor Business Management, Landscape Architecture</li> </ul>                                                                                                                                          |             |
| <b>Computer Science &amp; Mathematics Coursework</b> • Georgia Southern University • Statesboro, Georgia                                                                                                                                                                              | <b>2012</b> |
| <ul style="list-style-type: none"> <li>• Visual Basic, Calculus, C, C++, R, Statistics, Data Structures, Programming Principles, Linear Regression</li> </ul>                                                                                                                         |             |

## CERTIFICATIONS

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|-------------------------------------------------------|----------------------------------------------------|
| <b>2025</b> • Gauntlet AI Engineering Fellowship (G3) | <b>2020</b> • AutoML Pipelines (DataRobot Academy) |
| <b>2024</b> • NASA Space Apps Hackathon Finalist      | <b>2019</b> • Google Cloud Platform GCP Developer  |
| <b>2022</b> • Blockchain Hackathon Participant        | <b>2019</b> • AWS Microservices Cloud Practitioner |
| <b>2020</b> • TensorFlow Computer Vision              | <b>2019</b> • Y Combinator Startup School          |